

Selective Epistemic Deferral for Evidence-Grounded Multimodal Fact Checking

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Abstract

Multimodal fact-checking systems often treat every retrieved evidence item and every expert as equally useful. This is problematic when retrieval contains distractors or when a verifier is overconfident despite insufficient evidence. We study this failure mode on MOCHEG, a multimodal fact-checking benchmark with textual and visual evidence, while using a controlled text-evidence protocol so that retrieval and verification effects can be isolated. We introduce GraphCURE-B18B, a heterogeneous dual-expert verifier with asymmetric epistemic deferral: a direct verdict expert handles ordinary decisions, while a grounded-rationale expert can override the decision with “not enough information” when its estimated evidence sufficiency is high. The policy is selected on validation only and frozen before test evaluation. On the frozen validation split, B18B improves Macro-F1 from 0.6920 to 0.7112 (+1.92 points), with 10,000-sample paired bootstrap $P(\Delta > 0) = 0.9975$. Subgroup analysis shows that gains concentrate on rank-one evidence and NEI claims, while retrieval misses remain a failure mode. A secondary disagreement-aware distillation model, B19, compresses the complementary experts into one model and reaches 0.5494 Macro-F1 on the raw official test split at roughly eight times higher throughput than the eight-member B18B ensemble. We report official and strict test protocols separately and identify unresolved artifact inconsistencies explicitly.

1 Introduction

Automated fact checking is commonly decomposed into evidence retrieval and claim verification (Guo et al., 2022; Thorne et al., 2018). Modern systems increasingly retrieve heterogeneous evidence and use large language models, but two assumptions remain fragile: that all retrieved passages should be consumed by the verifier, and that an

expert trained for verdict prediction is also calibrated about evidence sufficiency. These assumptions are particularly consequential for multimodal fact checking, where evidence may be noisy, missing, duplicated, or distributed across modalities.

MOCHEG (Yao et al., 2023) provides a useful setting because it couples three-way truthfulness labels with retrieved textual and visual evidence and explanations. Existing MOCHEG systems model cross-modal interactions directly, for example with hypergraph transformers (Pang et al., 2025). However, comparisons are difficult when systems use different retrieval assumptions. We therefore first establish a fixed-corpus, text-retrieval protocol and then study the verifier and routing decisions under identical evidence.

Our central observation is that two verifiers have complementary error profiles. A direct verifier is strong when evidence appears decisive but can overcommit under missing evidence. A grounded expert, trained with structured explanations, is more sensitive to insufficiency but is not uniformly better. We turn this complementarity into a selective policy rather than averaging the experts. The direct expert remains the default; only a high-confidence grounded NEI signal can defer the decision.

Our contributions are:

- a reproducible, protocol-separated study of evidence-grounded verification on MOCHEG;
- GraphCURE-B18B, an asymmetric epistemic-deferral policy combining direct and grounded experts;
- a complete component, evidence-count, subgroup, and efficiency analysis; and
- B19, a secondary disagreement-aware distillation route that trades ensemble compute for a single student model.

We do not claim that every number in the internal development history is directly comparable. Official-test claims are restricted to frozen, auditable artifacts; unresolved discrepancies are marked in Section 10.

2 Related Work

Fact verification. FEVER introduced evidence-supported three-way verification at scale (Thorne et al., 2018); LIAR explored short political statements and fine-grained truthfulness (Wang, 2017); MultiFC expanded fact checking across fact-checking sources (Augenstein et al., 2019). Surveys describe retrieval, evidence selection, and verification as coupled but distinct stages (Guo et al., 2022).

Retrieval and reasoning. Recent work emphasizes that retrieval quality can dominate verification quality. RAV combines hybrid retrieval with joint verification (Zheng et al., 2024), while RAFTS synthesizes contrasting arguments from retrieved evidence (Yue et al., 2024). Realistic open-domain pipelines additionally decompose claims and retrieve evidence from the wild (Chen et al., 2024). These systems motivate our strict separation between fixed-corpus experiments and open-web experiments.

Multimodal fact checking. MOCHEG introduced end-to-end multimodal fact checking and explanation generation (Yao et al., 2023). HGTMFC uses hypergraph and line-graph propagation to model high-order interactions between claims and multimodal evidence (Pang et al., 2025). AMuFC argues that visual evidence should be used adaptively rather than universally (Jung et al., 2026). Our work shares the adaptive-evidence motivation, but studies an epistemic routing problem between verifier experts under a fixed textual retrieval protocol.

3 Task Definition and Protocols

Given a claim c and retrieved evidence $E = \{e_j\}_{j=1}^K$, predict $y \in \{\text{SUPPORTED}, \text{REFUTED}, \text{NEI}\}$. The principal P1 protocol uses MOCHEG’s fixed corpus and upstream text retrieval; it is not an open-web evaluation. The raw official test contains 2,442 claims. A strict robustness track removes eight verbatim cross-split duplicates and contains

2,434 claims. We report both tracks separately. Development validation contains 1,456 claims.

All routing thresholds are selected on validation or train-only out-of-fold data. Test labels are never used for selecting a seed, threshold, or policy in the confirmatory result. Open-web experiments in Phase C are not mixed into P1 tables.

4 Method

4.1 Direct and grounded experts

The direct expert f_D receives the claim and the top- K retrieved passages and predicts a distribution $p_D(y \mid c, E)$. It is trained with verdict cross-entropy. The grounded expert f_G uses the same evidence but is additionally trained from structured teacher explanations containing a verdict, key evidence identifiers, and missing-information fields. At inference it emits only the three-way verdict distribution; explanation generation is not required.

4.2 Asymmetric epistemic deferral

The policy uses the direct expert as an anchor:

$$\hat{y} = \begin{cases} \text{NEI}, & \arg \max_y p_G(y) = \text{NEI} \wedge p_G(\text{NEI}) \geq \tau, \\ \arg \max_y p_D(y), & \text{otherwise.} \end{cases} \quad (1)$$

We use $\tau = 0.49$, selected on validation and frozen thereafter. This rule is intentionally asymmetric: the grounded expert is not permitted to overwrite a direct Supported or Refuted decision with another decisive class. It can only defer when it detects insufficiency. This is different from posterior averaging, confidence routing, or an oracle router.

4.3 Disagreement-aware distillation

B19 is a secondary single-model variant. Let q_D and q_G be frozen teacher distributions. The student minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{KD}} T^2 \text{KL}(q \parallel p_S) + \lambda_{\text{cf}} \mathcal{L}_{\text{cf}}, \quad (2)$$

where q selects the correct teacher on teacher disagreement and averages the teachers when both agree. A counterfactual loss compares evidence sets with and without teacher-identified key passages. This model is reported as a compute-efficient secondary contribution, not as a replacement for the main B18B routing analysis.

5 Experimental Setup

5.1 Data and preprocessing

MOCHEG contains claims from PolitiFact and Snopes with text, image, and social evidence

Claim + retrieved passages $\rightarrow$ Direct verdict expert f_D ; the same input $\rightarrow$ Grounded sufficiency expert f_G . If f_G predicts NEI with probability at least τ , defer to NEI; otherwise retain f_D .

Figure 1: GraphCURE-B18B inference policy. The grounded branch is a selective epistemic sensor, not a symmetric replacement classifier.

Split/protocol	Claims	Labels used for selection	Role
Train	11,631	train only	optimization
Validation	1,456	validation	frozen policy selection
Official test	2,442	no	primary benchmark
Strict test	2,434	no	duplicate-robustness

Table 1: Protocol sizes. Official and strict test results are not pooled.

(Yao et al., 2023). Our P1 experiments use the project manifest, fixed text corpus, and Qwen3 dense/reranked retrieval. We retain upstream retrieval order and use $K = 5$ passages unless an evidence ablation changes K . No gold evidence is injected at validation or test.

5.2 Models and metrics

All verifiers use Qwen3-4B-Instruct-2507 with LoRA adapters. The B1 anchor uses five seeds; B18A uses grounded rationale distillation; B18B combines the frozen direct and grounded predictions. We report accuracy, class F1, Macro-F1, ECE when available, paired bootstrap intervals, and exact McNemar tests. Bootstrap samples are drawn at the claim level with 10,000 iterations.

5.3 Baselines and comparison discipline

Baselines include direct-only, grounded-only, equal posterior averaging, symmetric confidence routing, no-asymmetric-NEI routing, and oracle routing. Literature scores are reported in a separate protocol-aware table. MOCHEG, HGTMF, AMuFC, RAV, RAFTS, and open-web systems do not necessarily share evidence, split, or metric definitions; therefore a row is not treated as a direct leaderboard comparison unless its protocol is explicitly aligned.

6 Main Results

6.1 Frozen validation and ablations

Broader prior-work inventory. Table 3 lists more than ten relevant systems and benchmarks. The inventory is intentionally protocol-aware: a dash means that the source does not report a score

Variant	Accuracy	Macro-F1	Δ vs A8
A0 Direct-only	.700549	.692045	-.019160
A1 Grounded-only	.706731	.701191	-.010014
A2 Equal ensemble	.699863	.692773	-.018432
A3 Symmetric confidence	.698489	.691095	-.020110
A4 No asymmetric NEI	.710165	.704436	-.006769
A5 Disagreement-only	.706731	.701191	-.010014
A6 Max-confidence-only	.698489	.691095	-.020110
A8 Full B18B	.714973	.711205	0

Table 2: Frozen validation component ablations. A8 is the registered asymmetric policy with $\tau = .49$.

Work	Task/data	Reported result	Comparable to P1?
FEVER	fact verification	F1 (official)	No: Wikipedia evidence
LIAR	political claims	accuracy/F1	No: short statements
MultiFC	multi-source FC	macro-F1	No: source-specific setup
UKP-Athene	FEVER	F1	No: different evidence
MOCHEG	multimodal FC	F1=.4406	Closest benchmark
HGTMF	MOCHEG	F1=.4678	Same data; protocol differs
RAV	open-domain FC	retrieval/FC metrics	No: open retrieval
RAFTS	evidence reasoning	FC metrics	No: synthetic contrasts
Complex Claim Verification	open-domain claims	FC metrics	No: decomposed claims
AMuFC	multimodal FC	F1=.5400/.5460	Setting differs
GraphCURE-B1	MOCHEG P1	MF1=.5453	Aligned raw P1
GraphCURE-B18A	MOCHEG P1	MF1=.5551	Aligned raw P1

Table 3: Broader positioning inventory. Scores are reproduced only when the cited source or frozen project artifact reports them; otherwise the result is marked as not directly comparable.

that is directly comparable to our P1 fixed-corpus setting. Thus the table provides positioning, not a merged leaderboard.

B18B is the strongest registered validation variant. Relative to direct-only it gains 1.916 Macro-F1 points and 1.442 accuracy points. Removing the asymmetric NEI rule reduces Macro-F1 by 0.677 points, whereas equal averaging loses 1.843 points. These comparisons support the claim that the gain is due to the decision policy rather than simply adding a second model.

6.2 Official test results and leakage control

The frozen B18A heterogeneous ensemble reaches Macro-F1 0.55507 and accuracy 0.57535 on raw P1 official test, compared with the B1 five-seed baseline of 0.54531 and 0.56798. The paired bootstrap probability of improvement is 0.9839 with CI [+0.00097, +0.01850]. B19 disagreement distillation reaches Macro-F1 0.54940 with one student model. These numbers were produced after validation-only freezing.

The test-peak $\tau = 0.60$ result (Macro-F1 0.56166) is exploratory and is not used as the main

System	Macro-F1	Accuracy	Protocol
AMuFC	.5400	.5460	raw P1, source Table 3
HGTMFC	.4678	.4861	raw P1
B1 five-seed	.5453	.5680	raw P1
B18A super-ensemble	.5551	.5754	raw P1
B19 DKD seed 87	.5494	.5676	raw P1
B18B	[TODO: canonicalize]	[TODO: canonicalize]	strict/raw artifact

Table 4: Protocol-aware external comparison. Only rows with verified aligned settings should be interpreted as a direct comparison. B18B test values require reconciliation of two project artifacts before submission.

claim because the threshold was identified after inspecting test outcomes. Similarly, any B18B row whose canonical prediction file is not yet reconciled remains TODO.

7 Efficiency and Robustness Analysis

7.1 Inference efficiency

On an RTX 5090 with 256 common validation claims, batch size 4, and two warm-up batches, direct, grounded, B18B, and B19 systems require 320.72, 193.17, 513.88, and 64.47 ms per claim, respectively. B19 is approximately 7.97 times faster than B18B. All ensemble timings are sequential; peak allocated memory is therefore a per-checkpoint peak, not a simultaneous eight-model memory requirement.

7.2 Subgroup behavior

On the frozen validation audit, B18B improves Macro-F1 by 1.916 points, with 58 helpful and 37 harmful changes. The gain is largest for NEI (+3.239), rank-one evidence (+2.772), evidence hits (+2.280), and Snopes (+1.994). Retrieval misses decrease by 3.223 points and supported claims decrease by 1.404 points. These results suggest that the method is an evidence-sufficiency correction, not a universal replacement for direct verification.

7.3 B19 distillation

Five-fold train-only OOF confirmation gives B19 control Macro-F1 0.660788 and disagreement-KD Macro-F1 0.667607, a +0.006819 difference. Four of five folds improve and paired bootstrap $P(\Delta > 0) = 0.9517$. This supports B19 as a practical compression of complementary expertise, while B18B remains the main routing contribution.

8 Error Analysis

The dominant helpful transition is direct Refuted/Supported to grounded NEI on gold NEI

claims. Harmful transitions are direct Supported to NEI (28 cases) and direct Refuted to NEI (9 cases). This asymmetry explains why B18B improves NEI while slightly reducing decisive-class F1. The failure atlas also shows that routing under missing or rank-2–5 evidence is less reliable. A future system should estimate retrieval reliability before allowing deferral, but using gold rank or QREL status at inference would be invalid.

9 Discussion

The results support a narrow interpretation of the contribution. Evidence-grounded explanations are useful as a sufficiency sensor, but explanation supervision alone is not consistently beneficial: a matched explanation-loss ablation produced approximately zero aggregate gain. The useful object is the asymmetric interface between epistemic sensing and verdict prediction. This interface also exposes a compute–quality trade-off: B18B uses eight sequential expert passes, whereas B19 compresses the complementarity into one student.

10 Limitations and Ethical Considerations

MOCHEG is a single benchmark with two principal sources and potentially duplicated claims; we therefore report raw and strict tracks. The fixed-corpus protocol does not measure live web robustness. The grounded expert inherits teacher errors and pseudo-label noise. Subgroup analyses are descriptive and do not justify post-hoc policy tuning. The current repository contains inconsistent B18B test summaries; the final paper must replace the TODO row with a checksum-matched canonical evaluation. Automated fact checking can amplify source and retrieval bias; predictions should be treated as decision support, not authoritative adjudication.

11 Conclusion

We presented GraphCURE-B18B, a dual-expert asymmetric epistemic-deferral policy for evidence-grounded fact checking. On frozen validation it improves Macro-F1 by 1.92 points with strong paired-bootstrap evidence, and subgroup analysis attributes the gain to correcting NEI decisions when high-quality evidence is available. B19 demonstrates that complementary experts can be compressed into a single efficient student. The remaining priority is canonicalizing the B18B official-test

artifact and evaluating frozen policies on an external dataset before making a broad cross-domain claim.

Acknowledgments

[TODO: Add authors, affiliations, funding, and reproducibility URL after unblinding.]

References

- Isabelle Augenstein, Christina Lioma, Dongsheng Wang, Lucas Lima, Casper Hansen, Christian Hansen, and Jakob Simonsen. 2019. [Multifc: A real-world multi-domain dataset for evidence-based fact checking of claims](#). In *Proceedings of EMNLP-IJCNLP*, pages 4684–4696.
- Jifan Chen, Grace Kim, Aniruddh Sriram, Greg Durrett, and Eunsol Choi. 2024. [Complex claim verification with evidence retrieved in the wild](#). In *Proceedings of NAACL-HLT*, pages 3569–3587.
- Zhijiang Guo, Michael Schlichtkrull, and Andreas Vlachos. 2022. [A survey on automated fact-checking](#). *Transactions of the ACL*, 10:178–206.
- Jaeyoon Jung, Yejun Yoon, and Kunwoo Park. 2026. Is a picture worth a thousand words? adaptive multi-modal fact-checking with visual evidence necessity. *arXiv preprint arXiv:2604.04692*.
- Hui Pang, Chaozhuo Li, Litian Zhang, Senzhang Wang, and Xi Zhang. 2025. [Beyond text: Fine-grained multi-modal fact verification with hypergraph transformers](#). In *Proceedings of AAAI*, volume 39, pages 6389–6397.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. [Fever: A large-scale dataset for fact extraction and verification](#). In *Proceedings of NAACL-HLT*, pages 809–819.
- William Yang Wang. 2017. [Liar, liar pants on fire: A new benchmark dataset for fake news detection](#). In *Proceedings of ACL*, pages 422–426.
- Barry Menglong Yao, Aditya Shah, Lichao Sun, Jin-Hee Cho, and Lifu Huang. 2023. [End-to-end multimodal fact-checking and explanation generation: A challenging dataset and models](#). In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2733–2743.
- Zhenrui Yue, Huimin Zeng, Lanyu Shang, Yifan Liu, Yang Zhang, and Dong Wang. 2024. Retrieval augmented fact verification by synthesizing contrastive arguments. In *Proceedings of ACL*.
- Liwen Zheng, Chaozhuo Li, Xi Zhang, Yuming Shang, Feiran Huang, and Haoran Jia. 2024. Evidence retrieval is almost all you need for fact verification. In *Findings of ACL*, pages 9274–9281.

A Reproducibility Checklist

The repository records manifests, retrieval signatures, prediction files, training summaries, seeds, and protocol flags. The main scripts are ‘train_mocheg_qwen3_lora_verifier.py’, ‘analyze_mocheg_b18b_component_ablations.py’, ‘analyze_mocheg_b18b_subgroups.py’, and ‘benchmark_mocheg_b18b_efficiency.py’. Before submission, add exact GPU driver versions, package lock information, canonical B18B test hashes, and author metadata.